

Module-V

Numerical Solution of Ordinary Differential Equations

Let $\frac{dy}{dx} = f(x, y)$ is a differential equation whose solution is $y = f(x)$ and $y(x_0) = y_0$ be initial conditions.

Taylor's series method:

$$y = f(x) = y_0 + \frac{(x - x_0)}{1!}y'_0 + \frac{(x - x_0)^2}{2!}y''_0 + \frac{(x - x_0)^3}{3!}y'''_0 + \dots$$

Problem 5.1.1

Find by Taylor's series method the value of y at $x = 0.1$ and $x = 0.2$ to five places of decimal places from

$$\frac{dy}{dx} = x^2y - 1, y(0) = 1$$

Solution:

Given

$$y' = \frac{dy}{dx} = x^2y - 1$$
$$y(0) = 1$$

Also know, $y(x_0) = y_0$

$\therefore x_0 = 0, y_0 = 1$

As we know that, Taylor's method to find $y(x)$ is given by

$$y = f(x) = y_0 + \frac{(x - x_0)}{1!}y'_0 + \frac{(x - x_0)^2}{2!}y''_0 + \frac{(x - x_0)^3}{3!}y'''_0 + \dots$$



From the initial condition, $x_0 = 0, y_0 = 1$.

Given that

$$y' = x^2 y - 1 \quad (2)$$

At initial,

$$\begin{aligned} y'_0 &= x_0^2 y_0 - 1 \\ &= 0^2(1) - 1 = -1 \\ y'_0 &= -1 \end{aligned}$$

Find the derivative of equation (2),

$$y'' = x^2 y' + y \cdot 2x - 0 \quad (3)$$

At initial,

$$\begin{aligned} y''_0 &= x_0^2 y'_0 + y_0 \cdot 2x_0 - 0 \\ &= 0^2(-1) + 1(2)(0) - 0 \\ y''_0 &= 0 \end{aligned}$$

Find the derivative of equation (3),

$$y''' = x^2 y'' + y'.2x + 2y + 2xy' \quad (4)$$

At initial,

$$\begin{aligned} y_0''' &= x_0^2 y_0'' + y_0'.2x_0 + 2y_0 + 2x_0 y_0' \\ &= (0)^2(0) + (-1).2(0) + 2(1) + 2(0)(-1) \\ &= 0 + 0 + 2(1) + 0 = 2 \\ y_0''' &= 2 \end{aligned}$$

Find the derivative of equation (4),

$$\begin{aligned} y'''' &= (x^2 y''' + 2xy'') + (2y'(1) + 2xy'') + 2y' + (2(1)y' + 2xy'') \\ y'''' &= x^2 y''' + 6xy'' + 6y' \end{aligned}$$

At initial,

$$\begin{aligned} y_0'''' &= x_0^2 y_0''' + 6x_0 y_0'' + 6y_0' \\ &= (0)^2(2) + 6(0)(0) + 6(-1) = 0 + 0 + 6(-1) = -6 \\ y_0'''' &= -6 \end{aligned}$$

Substitute these values in the formula at equation (1), we get

$$\begin{aligned}y(x) &= y_0 + (x-0)(y'_0) + \frac{(x-0)^2}{2!}(y''_0) + \frac{(x-0)^3}{3!}y'''_0 + \frac{(x-0)^4}{4!}y''''_0 \\&= 1 + (x-0)(-1) + \frac{(x-0)^2}{2!}(0) + \frac{(x-0)^3}{3!}(2) + \frac{(x-0)^4}{4!}(-6) = 1\end{aligned}$$

$$y(x) = 1 - x + \frac{x^3}{3} - \frac{x^4}{4}$$

$$\begin{aligned}y(0.1) &= 1 - (0.1) + \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4} \\&\approx 0.90030\end{aligned}$$

$$\begin{aligned}y(0.2) &= 1 - (0.2) + \frac{(0.2)^3}{3} - \frac{(0.2)^4}{4} \\&\approx 0.80226\end{aligned}$$

Problem 5.1.2

Solve the given equation for $y(1.1)$ using Taylor's series method.

$$\frac{dy}{dx} = 2y + 3e^x, \quad y(0) = 0,$$

Given,

$$\begin{aligned}y' &= \frac{dy}{dx} = 2y + 3e^x \\y(0) &= 0 \\ \Rightarrow y(x_0) &= y_0 \\ \Rightarrow x_0 &= 0, \quad y_0 = 0.\end{aligned}$$

From the Taylor's series, we have

$$y = f(x) = y_0 + \frac{(x - x_0)}{1!}y'_0 + \frac{(x - x_0)^2}{2!}y''_0 + \frac{(x - x_0)^3}{3!}y'''_0 + \dots$$

From the initial condition, $x_0 = 0$, $y_0 = 0$.

Given that,

$$y' = 2y + 3e^x \quad (6)$$

At initial,

$$y'_0 = 2y_0 + 3e^{x_0} = 2(0) + 3e^0 = 0 + 3(1) = 3$$

$$y'_0 = 3$$

Find the derivative of equation (6),

$$y'' = 2y' + 3e^x \quad (7)$$

At initial,

$$\begin{aligned} y''_0 &= 2y'_0 + 3e^{x_0} \\ &= 2(3) + 3e^0 = 6 + 3(1) \end{aligned}$$

$$y''_0 = 9$$

Find the derivative of equation (7),

$$y''' = 2y'' + 3e^x \quad (8)$$

At initial,

$$y_0''' = 2y_0'' + 3e^{x_0} = 2(9) + 3e^0 = 21$$

$$y_0''' = 21$$

Find the derivative of equation (8),

$$y'''' = 2y''' + 3e^x$$

At initial,

$$y_0'''' = 2y_0''' + 3e^{x_0} = 2(21) + 3e^0$$

$$y_0'''' = 45$$

Substituting these values in equation (5), we get

$$y = f(x) = y_0 + \frac{(x - x_0)}{1!} y'_0 + \frac{(x - x_0)^2}{2!} y''_0 + \frac{(x - x_0)^3}{3!} y'''_0 + \frac{(x - x_0)^4}{4!} y''''_0 + \dots$$

$$y(x) = 0 + \frac{(x - 0)}{1!} (3) + \frac{(x - 0)^2}{2!} (9) + \frac{(x - 0)^3}{3!} (21) + \frac{(x - 0)^4}{4!} (45)$$

$$y(x) = 3x + \frac{x^2}{2} (9) + \frac{x^3}{6} (21) + \frac{x^4}{24} (45)$$

$$\begin{aligned} y(1.1) &= 3(1.1) + \frac{(1.1)^2}{2} (9) + \frac{(1.1)^3}{6} (21) + \frac{(1.1)^4}{24} (45) \\ &= 3.3 + 5.445 + 4.6585 + 2.7452 = 16.1457 \end{aligned}$$

Problem 5.1.3

From Taylor's series method, find $y(0.1)$ considering up to fourth degree term if $y(x)$ satisfies the equation $\frac{dy}{dx} = x - y^2$, $y(0) = 1$.

Taylor's series expansion is given by

$$y(x) = y_0 + (x - x_0)y'_0 + \frac{(x - x_0)^2}{2!}y''_0 + \frac{(x - x_0)^3}{3!}y'''_0 + \frac{(x - x_0)^4}{4!}y''''_0 + \dots \quad (9)$$

From initial condition $y(0) = 1$, we can conclude $x_0 = 0 \Rightarrow y_0 = 1$.

Given equation can be written as $y' = x - y^2$, Therefore, we substitute the values in (9), we have

$$y(x) = 1 + (x - 0)^2 y'(0) + \frac{(x - 0)^2}{2!} y''(0) + \frac{(x - 0)^3}{3!} y'''(0) + \frac{(x - 0)^4}{4!} y''''(0) \quad (10)$$

Given,

$$y' = x - y^2 \quad (12)$$

At initial,

$$y'_0 = 0 - 1^2 = -1$$

Find the derivative of equation (12)

$$y'' = 1 - 2yy' \quad (13)$$

At initial,

$$y''_0 = 1 - 2(1)(-1) = 3$$

Find the derivative of equation (13)

$$y''' = 0 - 2 [yy'' + (y')^2] \quad (14)$$

At initial,

$$y'''_0 = -2 [(1)(3) + (-1)^2] = -8$$

Find the derivative of equation (14)

$$y'''' = -2 [yy''' + y'y'' + 2y'y''] = -2 [yy'' + 3y'y''] \quad (15)$$

At initial,

$$y_0'''' = -2 [(1)(-8) + 3(-1)(3)] = 34$$

$y(x) \Rightarrow y(0.1)$ Substitute y', y'', y''', y'''' in equation (9). Therefore

$$y(0.1) = 1 + (0.1)(-1) + \frac{(0.1)^2}{2}(3) + \frac{(0.1)^3}{6}(-8) + \frac{(0.1)^4}{24}(34)$$

$$y(0.1) = 0.9138$$

Practice Problem

- 1 Consider the first order differential equation $\frac{dy}{dx} = x + y$, with initial condition $y(1) = 0$. Find y value at $x = 1.2$.
- 2 Use Taylor's series method to approximate y when $x = 0.1$, convert to 4 decimal places given that $\frac{dy}{dx} = 3x + y^2$ and $y = 1$ when $x = 0$ by taking the first five terms of Taylor's series expansions.

Given $y' = \frac{dy}{dx} = f(x, y)$ and initial condition $y(x_0) = y_0$.
Euler's method formula

$$y_{i+1} = y_i + hf(x_i, y_i)$$

Problem 5.2.1

Using Euler's method solve for y at $x = 0.1$ from $\frac{dy}{dx} = x + y + xy$, $y(0) = 1$ taking $h = 0.025$

Given $h = 0.025$

$$x_0 = 0$$

$$x_1 = x_0 + h = 0 + 0.025 = 0.025$$

$$x_2 = x_0 + 2h = 0 + 2(0.025) = 0.05$$

$$x_3 = x_0 + 3h = 0 + 3(0.025) = 0.075$$

$$x_4 = x_0 + 4h = 0 + 4(0.025) = 0.1$$

Given

$$y' = \frac{dy}{dx} = x + y + xy$$

$$y(0) = 1 \Rightarrow y(x_0) = y_0$$

$$\therefore x_0 = 0, y_0 = 1$$

Also given, $h = 0.025$

Now, we can calculate y_1 using Euler formula

$$y_{i+1} = y_i + hf(x_i, y_i)$$

$$y_1 = y_0 + hf(x_0, y_0)$$

$$y_1 = 1 + 0.025 (x_0 + y_0 + x_0 y_0)$$

$$y_1 = 1 + 0.025 (0 + 1 + 0)$$

$$y_1 = y(x_1) = 1 + 0.025$$

$$i.e) y(0.025) = 1.025$$

$$x_1 = 0.025 \quad y_1 = 1.025$$

$$y_2 = y_1 + hf(x_1, y_1) = y_1 + h(x_1 + y_1 + x_1 y_1)$$

$$y_2 = 1.025 + 0.025 (0.025 + 1.025 + 0.025(1.025))$$

$$y_2 = 1.025 + 0.025 (1.0756)$$

$$\therefore y(0.05) = y(x_2) = 1.05189$$

$$y_3 = y_2 + hf(x_2, y_2)$$

$$y_3 = y_2 + h(x_2 + y_2 + x_2 y_2)$$

$$y_3 = y(x_3) = 1.05189 + 0.025 (0.05 + 1.05189 + 0.05(1.05189))$$

$$\therefore y(0.075) = 1.08075$$

$$x_3 = 0.075, y_3 = 1.08075$$

$$y_4 = y_3 + hf(x_3, y_3)$$

$$y_4 = y_3 + h(x_3 + y_3 + x_3 y_3)$$

$$y_4 = 1.08075 + 0.025 (0.075 + 1.08075 + 0.075(1.08075))$$

$$y_4 = y(x_4) = 1.11167$$

$$\therefore y(0.1) = 1.11167$$

Example 5.2.2

Using Euler's method solve for y at $x = 1$ from $\frac{dy}{dx} = 2e^x + y^2$, $y(0) = \frac{1}{2}$ taking $h = 0.25$.

Given $h = 0.25$

$$x_0 = 0$$

$$x_1 = x_0 + h = 0 + 0.25 = 0.25$$

$$x_2 = x_0 + 2h = 0 + 2(0.25) = 0.50$$

$$x_3 = x_0 + 3h = 0 + 3(0.25) = 0.75$$

$$x_4 = x_0 + 4h = 0 + 4(0.25) = 1$$

Given

$$y' = \frac{dy}{dx} = 2e^x + y^2$$

$$y(0) = \frac{1}{2} \Rightarrow y(x_0) = y_0$$

$$\therefore x_0 = 0, y_0 = \frac{1}{2}$$

Also given, $h = 0.25$

Now, we can calculate y_1 using Euler formula

$$y_{i+1} = y_i + hf(x_i, y_i)$$

$$y_1 = y_0 + hf(x_0, y_0) = \frac{1}{2} + 0.25 (2e^{x_0} + y_0^2)$$

$$y_1 = \frac{1}{2} + 0.25 \left(2(1) + \left(\frac{1}{2} \right)^2 \right)$$

$$y_1 = y(x_1) = 1.0625$$

$$i.e) y(0.25) = 1.0625$$

$$\text{So, } x_1 = 0.25, y_1 = 1.0625$$

$$y_2 = y_1 + hf(x_1, y_1) = y_1 + h(2e^{x_1} + y_1^2)$$

$$y_2 = 1.0625 + 0.25 (2e^{0.25} + (1.0625)^2) = 1.0625 + 0.25 (3.697)$$

$$\therefore y(0.50) = y(x_2) = 1.9867$$

$$\text{So, } x_2 = 0.50, y_2 = 1.9867$$

$$y_3 = y_2 + hf(x_2, y_2) = y_2 + h(2e^{x_2} + y_2^2)$$

$$y_3 = 1.9867 + (0.25)(2e^{0.50} + 1.9867^2) = 1.9867 + (0.25)(7.2446)$$

$$y_3 = y(x_3) = 3.7979$$

$$\therefore y(0.75) = 3.7979$$

$$\text{So, } x_3 = 0.75, y_3 = 3.7979$$

$$y_4 = y_3 + hf(x_3, y_3) = y_3 + h(2e^{x_3} + y_3^2)$$

$$y_4 = 3.7979 + (0.25)(2e^{0.75} + 3.7979^2) = 3.7979 + (0.25)(18.6579)$$

$$y_4 = 8.4624$$

$$y_4 = y(x_4) = 8.4624$$

$$\therefore y(1) = 8.4624$$

Given

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0$$

Modified Euler's formula is given by

$$y_{n+1}^{(r+1)} = y_n + \frac{h}{2} \left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(r)}) \right]$$

where, $r = 0, 1, 2, \dots$ where

$$y_{n+1}^{(0)} = y_n + hf(x_n, y_n), n = 0, 1, 2, \dots,$$

using Euler's formula.

Example 5.3.1

Using Modified Euler's method find $y(0.2)$ given that $\frac{dy}{dx} = x + y$, $y(0) = 1$.
Correct to 4 decimal places.

Given

$$f(x, y) = x + y,$$

$$x_0 = 0, y_0 = 1,$$

$$x_1 = 0.2,$$

$$h = x_1 - x_0 = 0.2 - 0$$

$$h = 0.2$$

Initial approximation

$$y_{n+1}^{(0)} = y_n + hf(x_n, y_n)$$

Put $n = 0$

$$\begin{aligned} y_1^{(0)} &= y_0 + hf(x_0, y_0) \\ &= 1 + 0.2(0 + 1) \end{aligned}$$

$$y_1^{(0)} = 1.2$$

Modified Euler formula is

$$y_{n+1}^{(r+1)} = y_n + \frac{h}{2} \left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(r)}) \right] \quad (16)$$

First approximation:

Put $r = 0$ and $n = 0$ in equation (16)

$$\begin{aligned}y_1^{(1)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(0)})] \\&= 1 + \frac{0.2}{2} [(x_0 + y_0) + (x_1 + y_1^{(0)})] \\&= 1 + \frac{0.2}{2} [(0 + 1) + (0.2 + 1.2)] \\y_1^{(1)} &= 1.24\end{aligned}$$

Second Approximation

Put $r = 1$, $n = 0$ in equation (16)

$$\begin{aligned}y_1^{(2)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})] \\&= 1 + \frac{0.2}{2} [(0 + 1) + (0.2 + 1.24)] \\y_1^{(2)} &= 1.244\end{aligned}$$

Third approximation:

Put $r = 2$, $n = 0$ in equation (16)

$$\begin{aligned}y_1^{(3)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})] \\&= 1 + \frac{0.2}{2} [(0 + 1) + (0.2 + 1.244)] \\y_1^{(3)} &= 1.2444\end{aligned}$$

Fourth Approximation

Put $r = 3$, $n = 0$ in equation (16)

$$\begin{aligned}y_1^{(4)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(3)})] \\&= 1 + \frac{0.2}{2} [(0 + 1) + (0.2 + 1.2444)] \\y_1^{(4)} &= 1.24444\end{aligned}$$

Since, $y_1^{(3)}$ and $y_1^{(4)}$ are the same at corrected to four decimal places.

$$y_1 = y(x_1) = y(0.2) = 1.2444$$

Example 5.3.2

Use Modified Euler's Method to find the approximate value of $y(1.1)$ for the solution of the initial value problem $\frac{dy}{dx} = 2xy$, $y(1) = 1$ correct to 3 decimal places, perform 2 iterations.

$$f(x, y) = 2xy$$

$$x_0 = 1, y_0 = 1, x_1 = 1.1$$

$$h = x_1 - x_0 = 1.1 - 1 = 0.1$$

$$y(1.1) = 1.2355$$

Example 5.3.3

Find $y(1.2)$ and $y(1.4)$ by modified Euler's method given that $\frac{dy}{dx} = \frac{2y}{x} + x^3$, $y(1) = 0.5$ correct to 3 decimal places.

Given $f(x, y) = \frac{2y}{x} + x^3$

$$\begin{aligned}x_0 &= 1, & y_0 &= 0.5, \\ \text{and } x_1 &= 1.2, & x_2 &= 1.4\end{aligned}$$

$x_0 = 1$	$x_1 = x_0 + h = 1.2$	$x_2 = x_0 + 2h = 1.4$
$y_0 = 0.5$	$h = x_1 - x_0 = 0.2$	$h = \frac{x_2 - x_0}{2} = 0.2$

(i). To find $y(1.2)$

$$h = x_1 - x_0 = 1.2 - 1 = 0.2$$

Modified Euler's formula is given by

$$y_{n+1}^{(r+1)} = y_n + \frac{h}{2} \left[f(x_n, y_n) + f\left(x_{n+1}, y_{n+1}^{(r)}\right) \right] \quad (17)$$

where,

$$y_{n+1}^{(0)} = y_n + hf(x_n, y_n) \quad (18)$$

Step:1

To find $y_1 = y(x_1) = y(1.2)$

Put $n = 0$ in equation (17) and (18)

$$y_1^{(r+1)} = y_0 + \frac{h}{2} \left[f(x_0, y_0) + f(x_1, y_1^{(r)}) \right] \quad (19)$$

$$\text{where } y_1^{(0)} = y_0 + hf(x_0, y_0) \quad (20)$$

Initial approximation from equation (20)

$$\begin{aligned} y_1^{(0)} &= 0.5 + 0.2 \left[\frac{2y_0}{x_0} + (x_0)^3 \right] \\ &= 0.5 + 0.2 \left[\frac{(2)(0.5)}{1} + (1)^3 \right] \\ y_1^{(0)} &= 0.9 \end{aligned}$$

First Approximation, put $r = 0$ in equation (19)

$$\begin{aligned}y_1^{(1)} &= y_0 + \frac{h}{2} \left[f(x_0, y_0) + f(x_1, y_1^{(0)}) \right] \\&= 0.5 + \frac{0.2}{2} \left[\left(\frac{2y_0}{x_0} + (x_0)^3 \right) + \left(\frac{2y_1^{(0)}}{x_1} + (x_1)^3 \right) \right] \\y_1^{(1)} &= 1.0227\end{aligned}$$

Second Approximations, put $r = 1$ in equation (19)

$$\begin{aligned}y_1^{(2)} &= y_0 + \frac{h}{2} \left[f(x_0, y_0) + f(x_1, y_1^{(1)}) \right] \\&= 0.5 + \frac{0.2}{2} \left[\left(\frac{2y_0}{x_0} + (x_0)^3 \right) + \left(\frac{2y_1^{(1)}}{x_1} + (x_1)^3 \right) \right] \\y_1^{(2)} &= 1.043\end{aligned}$$

Similarly, $y_1^{(3)} = 1.046$ and $y_1^{(4)} = 1.046$.

Since $y_1^{(3)}$ and $y_1^{(4)}$ are the same correct to 4 decimal places

$$y(1.2) = 1.046 \quad (21)$$

Step: To find $y_2 = y(x_2) = y(1.4)$

Put $n = 1$ in equations (17) and (18)

$$y_2^{(r+1)} = y_1 + \frac{h}{2} \left[f(x_1, y_1) + f(x_2, y_2^{(r)}) \right] \quad (22)$$

$$y_2^{(0)} = y_1 + h(x_1, y_1) \quad (23)$$

Initial Approximation from equation 23

$$\begin{aligned} y_2^{(0)} &= 1.046 + 0.2 \left[\frac{2y_1}{x_1} + (x_1)^3 \right] \\ &= 1.046 + 0.2 \left[\frac{2(1.046)}{1.2} + (1.2)^3 \right] \\ y_2^{(0)} &= 1.740 \end{aligned}$$

First Approximation

Put $r = 0$ in equation (22)

$$\begin{aligned}y_2^{(1)} &= y_1 + \frac{h}{2} \left[f(x_1, y_1) + f(x_2, y_2^{(0)}) \right] \\&= 1.046 + \frac{0.2}{2} \left[\frac{2(1.046)}{1.2} + (1.2)^3 + \frac{2(1.74)}{1.4} + (1.4)^3 \right] \\y_2^{(1)} &= 1.916.\end{aligned}$$

Similarly,

$$\begin{aligned}y_2^{(2)} &= 1.941, & y_2^{(3)} &= 1.944 \\y_2^{(4)} &= 1.945, & y_2^{(5)} &= 1.945\end{aligned}$$

Since, $y_2^{(4)}$ and $y_2^{(5)}$ are the same correct to the decimal places

$$y(1.4) = 1.945$$

Runge-kutta method of 4th order

Given

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0$$

The Runge-kutta method of 4th order is given by

$$y_{n+1} = y_n + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

where

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

Example 5.4.1

Apply Runge Kutta Method of fourth order to find an approximate value of $y(0.1)$ and $y(0.2)$ of $\frac{dy}{dx} = x + y^2$, $y(0) = 1$, correct to three decimal places.

Given $f(x, y) = x + y^2$

$$x_0 = 0,$$

$$y_0 = 1$$

$$x_1 = 0.1,$$

$$x_2 = 0.2$$

$$h = x_1 - x_0 = 0.1 - 0 = 0.1$$

We have to calculate y_1 and y_2 .

The Runge-Kutta method of 4th order is given by

$$y_{n+1} = y_n + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4],$$

where,

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

Step 1: Finding $y(0.1)$ i.e. y_1

Put $n = 0$ in equation (24)

$$y_1 = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \quad (25)$$

where

$$\begin{aligned} k_1 &= hf(x_0, y_0) = h [x_0 + (y_0)^2] \\ &= 0.1 [0 + (1)^2] \end{aligned}$$

$$k_1 = 0.1$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$\begin{aligned} k_2 &= h \left[\left(x_0 + \frac{h}{2}\right) + \left(y_0 + \frac{k_1}{2}\right)^2 \right] \\ &= 0.1 \left[\left(0 + \frac{0.1}{2}\right) + \left(1 + \frac{0.1}{2}\right)^2 \right] \end{aligned}$$

$$k_2 = 0.1152$$

$$\begin{aligned}
 k_3 &= hf \left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2} \right) \\
 &= h \left[\left(x_0 + \frac{h}{2} \right) + \left(y_0 + \frac{k_2}{2} \right)^2 \right] \\
 &= 0.1 \left[\left(0 + \frac{0.1}{2} \right) + \left(1 + \frac{0.1152}{2} \right)^2 \right] \\
 k_3 &= 0.1168
 \end{aligned}$$

$$\begin{aligned}
 k_4 &= hf (x_0 + h, y_0 + k_3) \\
 &= h [(x_0 + h) + (y_0 + k_3)^2] \\
 &= 0.1 [(0 + 0.1) + (1 + 0.1168)^2] \\
 k_4 &= 0.1347
 \end{aligned}$$

Substitute y_0, k_1, k_2, k_3 and k_4 in equation (25), we get

$$y_1 = 1 + \frac{1}{6} [0.1 + 2(0.1152) + 2(0.1168) + 0.1347]$$

$$y_1 = 1.1164$$

$$\text{i.e., } y(0.1) = 1.1164$$

Step 2: Finding $y(0.2)$ i.e., y_2

Put $n = 1$ in equation (24), we get

$$y_2 = y_1 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

where,

$$k_1 = hf(x_1, y_1)$$

$$\begin{aligned} k_1 &= 0.1 [x_1 + y_1^2] \\ &= 0.1 [0.1 + (1.1164)^2] \end{aligned}$$

$$k_1 = 0.1346$$

$$\begin{aligned}
 k_2 &= hf \left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2} \right) \\
 &= h \left[\left(x_1 + \frac{h}{2} \right) + \left(y_1 + \frac{k_1}{2} \right)^2 \right] \\
 &= 0.1 \left[\left(0.1 + \frac{0.1}{2} \right) + \left(1.1164 + \frac{0.1346}{2} \right)^2 \right] \\
 k_2 &= 0.1551
 \end{aligned}$$

$$\begin{aligned}
 k_3 &= hf \left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2} \right) \\
 &= h \left[\left(x_1 + \frac{h}{2} \right) + \left(y_1 + \frac{k_2}{2} \right)^2 \right] \\
 &= 0.1 \left[\left(0.1 + \frac{0.1}{2} \right) + \left(1.1164 + \frac{0.1551}{2} \right)^2 \right]
 \end{aligned}$$

$$\begin{aligned}
 k_4 &= hf [x_1 + h, y_1 + k_3] \\
 &= h [(x_1 + h) + (y_1 + k_3)^2] \\
 &= 0.1 [(0.1 + 0.1) + (1.1164 + 0.1575)^2] \\
 k_4 &= 0.1822
 \end{aligned}$$

Substituting y_1, k_1, k_2, k_3 and k_4 in equation (24), we get

$$y_2 = 1.1164 + \frac{1}{6} [0.1346 + 2(0.1575) + 0.1822]$$

$$y_2 = 1.2734$$

$$\text{i.e., } y(0.2) = 1.2734.$$

Example 5.4.2

Use Runge-kutta method of fourth order to approximate y when $x = 0.1$, given that $y = 1$ when $x = 0$, $\frac{dy}{dx} = x + y$, correct to 4 decimal places.

Given,

$$f(x, y) = x + y$$

$$x_0 = 0, y_0 = 1, x_1 = 0.1$$

$$h = x_1 - x_0 = 0.1 - 0 = 0.1$$

$$y_1 = ? \Rightarrow y(0.1) = ?$$

The Runge-Kutta method of 4th order is given by

$$y_{n+1} = y_n + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \quad (26)$$

$$\text{where } k_1 = hf(x_n, y_n) \quad (27)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right) \quad (28)$$

$$k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

Put $n = 0$ in equation (26)

$$y_1 = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

where,

$$\begin{aligned} k_1 &= hf(x_0, y_0) \\ k_2 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \\ k_3 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \\ k_4 &= hf(x_0 + h, y_0 + k_3) \end{aligned} \tag{30}$$

$$\begin{aligned}\therefore k_1 &= 0.1 [x_0 + y_0] \\ &= 0.1 [0 + 1] = 0.1\end{aligned}$$

$$\begin{aligned}k_2 &= 0.1 \left[\left(x_0 + \frac{h}{2} \right) + \left(y_0 + \frac{k_1}{2} \right) \right] \\ &= 0.1 \left[\left(0 + \frac{0.1}{2} \right) + \left(1 + \frac{0.1}{2} \right) \right]\end{aligned}$$

$$k_2 = 0.11$$

$$\begin{aligned}k_3 &= 0.1 \left[\left(x_0 + \frac{h}{2} \right) + \left(y_0 + \frac{k_2}{2} \right) \right] \\ &= 0.1 \left[\left(0 + \frac{0.1}{2} \right) + \left(1 + \frac{0.11}{2} \right) \right]\end{aligned}$$

$$k_3 = 0.1105$$

$$\begin{aligned}k_4 &= 0.1 [(x_0 + h) + (y_0 + k_3)] \\ &= 0.1 [(0 + 0.1) + (1 + 0.1105)]\end{aligned}$$

$$k_4 = 0.1210$$

$$k_1 = 0.1, k_2 = 0.11, k_3 = 0.1105, k_4 = 0.1210, y_0 = 1.$$

Substitute all the values in equation (30), we get

$$y_1 = 1 + \frac{1}{6} [0.1 + 2(0.11) + 2(0.1105) + 0.1210]$$

$$y_1 = 1.1103$$

$$\text{i.e. } y(0.1) = 1.1103$$

Example 5.4.3

Use Runge-kutta method of fourth order to obtain an approximation to $y(1.5)$ for the solution of $\frac{dy}{dx} = 2xy$, $y(1) = 1$, correct to 4 decimal places.

Hint:

$$f(x, y) = 2xy, x_0 = 1, y_0 = 1$$

$$x_1 = 1.5$$

$$h = x_1 - x_0 = 1.5 - 1 = 0.5$$

$$y_1 = 3.4543$$

Adams- Bashforth Predictor-corrector method

Given

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0$$

Adams- Bashforth's predictor and corrector formula is given by

$$y_{4,p} = y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3) \quad (\text{Predictor Formula}) \quad (31)$$

$$y_{4,c}^{(r+1)} = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4^{(r)}) \quad (\text{Corrector Formula}) \quad (32)$$

where

$$\begin{aligned} f_1 &= f(x_1, y_1), & f_2 &= f(x_2, y_2), & f_3 &= f(x_3, y_3) \\ f_4^{(r)} &= f(x_4, y_4^{(r)}), & f_4^{(0)} &= f(x_4, y_{4,p}^{(0)}) \end{aligned}$$

where

$$y_4^{(0)} = y_{4,p}, \text{ for } r = 0, \quad y_4^{(r)} = y_{4,c}^{(r)}, \text{ for } r \neq 0.$$

Example 5.5.1

Solve the initial value problem $\frac{dy}{dx} = 1 + xy^2$, $y(0) = 1$ for $x = 0.4$ by Milne's predictor and corrector method correct to three decimal places, given that

x	0.1	0.2	0.3
y	1.105	1.223	1.355

Given $f(x, y) = 1 + xy^2$, $h = x_2 - x_1 = 0.1$

$$x_0 = 0 \quad x_1 = 0.1 \quad x_2 = 0.2 \quad x_3 = 0.3 \quad x_4 = 0.4$$

$$y_0 = 1 \quad y_1 = 1.105 \quad y_2 = 1.223 \quad y_3 = 1.355 \quad y_4 = ?$$

Milne's predictor formula is given by

$$y_{4,p} = y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3) \quad (33)$$

x_i	y_i	$f_i = f(x_i, y_i) = 1 + x_i y_i^2$
$x_1=0.1$	$y_1=1.105$	$f_1 = 1 + x_1 y_1^2$ $= 1 + (0.1)(1.105)^2$ $= 1.122$
$x_2=0.2$	$y_2=1.223$	$f_2 = 1 + x_2 y_2^2$ $= 1 + (0.2)(1.223)^2$ $= 1.299$
$x_3=0.3$	$y_3=1.355$	$f_3 = 1 + x_3 y_3^2$ $= 1 + (0.3)(1.355)^2$ $= 1.550$

Substituting all the values in equation (33) we get,

$$y_{4,p} = 1 + \left(\frac{4(0.1)}{3} \right) [2(1.122) - 1.299 + 2(1.550)]$$

$$y_{4,p} = 1.526$$

Milne's corrector formula is given by

$$y_{4,c}^{(r+1)} = y_2 + \left(\frac{h}{3} \right) (f_2 + 4f_3 + f_4^{(r)}) \quad (34)$$

where

$$f_4^{(r)} = f(x_4, y_4^{(r)})$$

$$y_4^{(0)} = y_{4,p}, \quad r = 0$$

$$y_4^{(r)} = y_{4,c}^{(r)}, \quad r \neq 0.$$

First improvement:

Put $r = 0$ in equation (34)

$$y_{4,c}^{(1)} = y_2 + \left(\frac{h}{3}\right) (f_2 + 4f_3 + f_4^{(0)})$$

where

$$\begin{aligned} f_4^{(0)} &= f(x_4, y_4^{(0)}) = f(x_4, y_{4,p}) \\ &= 1 + x_4 (y_{4,p})^2 \end{aligned}$$

$$f_4^{(0)} = 1 + (0.4)(1.526)^2 = 1.931$$

$$\therefore y_{4,c}^{(1)} = 1.223 + \left(\frac{0.1}{3}\right) (1.299 + 4(1.550) + 1.931)$$

$$y_{4,c}^{(1)} = 1.537$$

Second improvement:

Put $r = 1$ in equation (34)

$$y_{4,c}^{(2)} = y_2 + \left(\frac{h}{3}\right) (f_2 + 4f_3 + f_4^{(1)})$$

where

$$\begin{aligned} f_4^{(1)} &= f(x_4, y_4^{(1)}) \\ &= 1 + x_4 (y_{4,c}^{(1)})^2 \\ &= 1 + (0.4)(1.537)^2 \end{aligned}$$

$$f_4^{(1)} = 1.944$$

$$\therefore y_{4,c}^{(2)} = 1.223 + \left(\frac{0.1}{3}\right) (1.299 + 4(1.550 + 1.944))$$

$$y_{4,c}^{(2)} = 1.537$$

Since, $y_{4,c}^{(1)}$ and $y_{4,c}^{(2)}$ are the same up to three decimal places

$$y(0.4) = 1.537$$

Example 5.5.2

Solve the initial value problem $\frac{dy}{dx} = \frac{x+y}{2}$, $y(0) = 2$ for $x = 2$ by Milne's predictor and corrector method correct to three decimal places, given that

x	0.5	1	1.5
y	2.636	3.595	4.968

Sol: $y(2) = 6.8732$

Example 5.5.3

Solve the initial value problem $\frac{dy}{dx} = \frac{(1+x^2)y^2}{2}$, $y(0) = 1$ for $x = 0.4$ by Milne's predictor and corrector method correct to three decimal places, given that

x	0.1	0.2	0.3
y	1.06	1.12	1.21

Sol: $y(2) = 1.2797$

$$T(0, y) = 100^{\circ}\text{C} \quad (\text{Left edge})$$

$$T(4, y) = 100^{\circ}\text{C} \quad (\text{Right edge})$$

$$T(x, 0) = 0^{\circ}\text{C} \quad (\text{Bottom edge})$$

$$T(x, 4) = 0^{\circ}\text{C} \quad (\text{Top edge})$$

Explicit method - Bender-Schmidt method

Bender-Schmidt method is used to solve the heat equation and similar partial equations.

The general equation of heat equation is

$$\frac{\partial^2 u}{\partial x^2} - a \frac{\partial u}{\partial t} = 0; \text{ where } \begin{cases} x & \rightarrow \text{indicate distance} \\ y & \rightarrow \text{indicate time} \end{cases}$$

General formula of Bender-Schmidt explicit method is

$$u_{i,j+1} = \frac{1}{2} [u_{i-1,j} + u_{i+1,j}]$$

- ❶ $\lambda = \frac{k}{ah^2}$ assume $[\lambda = \frac{1}{2}]$
- ❷ $u_{xx} = au_t$
- ❸ $h = \Delta x$ and $k = \Delta t$

Example 5.6.1

Solve $\frac{\partial^2 u}{\partial x^2} = 2 \frac{\partial u}{\partial t}$ given $u(0, t) = 0$, $u(4, t) = 0$, $u(x, 0) = x(4 - x)$, taking $\Delta x = 1$ and $\Delta t = 1$. Find the value of u upto $t = 5$ using Bender-Schmidt's explicit finite difference Scheme.

Given:

$$\begin{aligned} u_{xx} &= 2u_t & \Rightarrow & a = 2 \\ h &= \Delta x = 1 & \text{and} & k = \Delta t = 1 \\ \lambda &= \frac{k}{ah^2} & \Rightarrow & \lambda = \frac{1}{2(1)^2} = \frac{1}{2} \end{aligned}$$

x varies from 0 to 4 with $h = 1$.

$x :$	0	1	2	3	4
$i :$	0	1	2	3	4

Find the value of u upto $t = 5$ with $k = \Delta(t) = 1$.

$t :$	0	1	2	3	4	5
$j :$	0	1	2	3	4	5

Using the boundary conditions $u(0, t) = 0$ and $u(4, t) = 0$, we can find

$$u(0, t) = 0 \quad \Rightarrow \quad u_{0,j} = 0 \quad \text{First Column}$$

$$u(4, t) = 0 \quad \Rightarrow \quad u_{4,j} = 0 \quad \text{Last Column}$$

Using initial condition $u(x, 0) = x(4-x)$, we can find first row value as follows,

$$u(0, 0) = 0$$

$$u(1, 0) = 1(4 - 1) = 3$$

$$u(2, 0) = 2(4 - 2) = 4$$

$$u(3, 0) = 3(4 - 3) = 3$$

$$u(4, 0) = 4(4 - 4) = 0$$

General formula of Bender-Schmidt explicit method is

$$u_{i,j+1} = \frac{1}{2} [u_{i-1,j} + u_{i+1,j}]$$

The values of $u_{i,j}$ are tabulated below:

		$x = 0$	$x = 1$	$x = 2$	$x = 3$	$x = 4$
	$\begin{array}{c} i \\ \diagdown \\ j \end{array}$	0	1	2	3	4
$t = 0$	0	0	3	4	3	0
$t = 1$	1	0				0
$t = 2$	2	0				0
$t = 3$	3	0				0
$t = 4$	4	0				0
$t = 5$	5	0				0

		$x = 0$	$x = 1$	$x = 2$	$x = 3$	$x = 4$
	$\begin{array}{c} i \\ \diagdown \\ j \end{array}$	0	1	2	3	4
$t = 0$	0	0	3	4	3	0
$t = 1$	1	0	2	3	2	0
$t = 2$	2	0	1.5	2	1.5	0
$t = 3$	3	0	1	1.5	1	0
$t = 4$	4	0	0.75	1	0.75	0
$t = 5$	5	0	0.5	0.75	0.50	0

Example 5.6.2

Solve $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ given $u(0, t) = 0$, $u(5, t) = 0$, $u(x, 0) = x^2(25 - x^2)$. Compute $u(x, t)$ upto $t = 5$ with $\Delta x = 1$ using Benders-Schmidt's method.

Ans: Since λ value is not given, let us take $\lambda = \frac{1}{2}$.

Given:

$$u_{xx} = u_t \quad \Rightarrow \quad a = 1$$

Since, $h = \Delta x = 1$

$$\frac{k}{ah^2} = \lambda \quad \Rightarrow \quad \frac{k}{1(1)^2} = \frac{1}{2} \quad \Rightarrow \quad k = \Delta(t) = 0.5$$

Using the boundary conditions

$$u(0, t) = 0 \quad \Rightarrow \quad u_{0,j} = 0 \quad \text{First Column}$$

$$u(5, t) = 0 \quad \Rightarrow \quad u_{5,j} = 0 \quad \text{Last Column}$$

x varies from 0 to 5 with $h = 1$.

$x :$	0	1	2	3	4	5
$i :$	0	1	2	3	4	5

Find the value of u upto $t = 5$ with $k = \Delta(t) = 0.5$.

$t :$	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
$j :$	0	1	2	3	4	5	6	7	8	9	10

Next,

$$u(x, 0) = x^2(25 - x^2)$$

The value of first row is given by,

$$u(0, 0) = 0^2(25 - 0^2) = 0$$

$$u(1, 0) = 1^2(25 - 1^2) = 24$$

$$u(2, 0) = 2^2(25 - 2^2) = 84$$

$$u(3, 0) = 3^2(25 - 3^2) = 144$$

$$u(4, 0) = 4^2(25 - 4^2) = 144$$

$$u(5, 0) = 5^2(25 - 5^2) = 0$$

General formula of Bender-Schmidt explicit method is

$$u_{i,j+1} = \frac{1}{2} [u_{i-1,j} + u_{i+1,j}]$$

The values of $u_{i,j}$ are tabulated below:

		$x = 0$	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$
	$i \backslash j$	0	1	2	3	4	5
$t = 0$	0	0	24	84	144	144	0
$t = 0.5$	1	0	42	84	114	72	0
$t = 1$	2	0	42	78	78	57	0
$t = 1.5$	3	0	39	60	67.5	39	0
$t = 2$	4	0	30	53.25	49.5	33.75	0
$t = 2.5$	5	0	26.625	39.75	43.5	24.75	0
$t = 3$	6	0	19.875	35.0625	37.25	21.75	0
$t = 3.5$	7	0	17.5313	26.0625	28.4063	16.125	0
$t = 4$	8	0	13.0313	22.9688	21.0938	14.2031	0
$t = 4.5$	9	0	11.4844	17.0625	18.5859	10.5469	0
$t = 5$	10	0	8.5313	15.0351	13.8047	9.2930	0